

EXPERIMENT 3:

Write program demonstrates how to perform morphological analysis using the NLTK library in Python.

AIM:

To perform morphological analysis using the NLTK library in Python and obtain the root form (stem) of words.

PROGRAM:

```
from nltk.stem import PorterStemmer

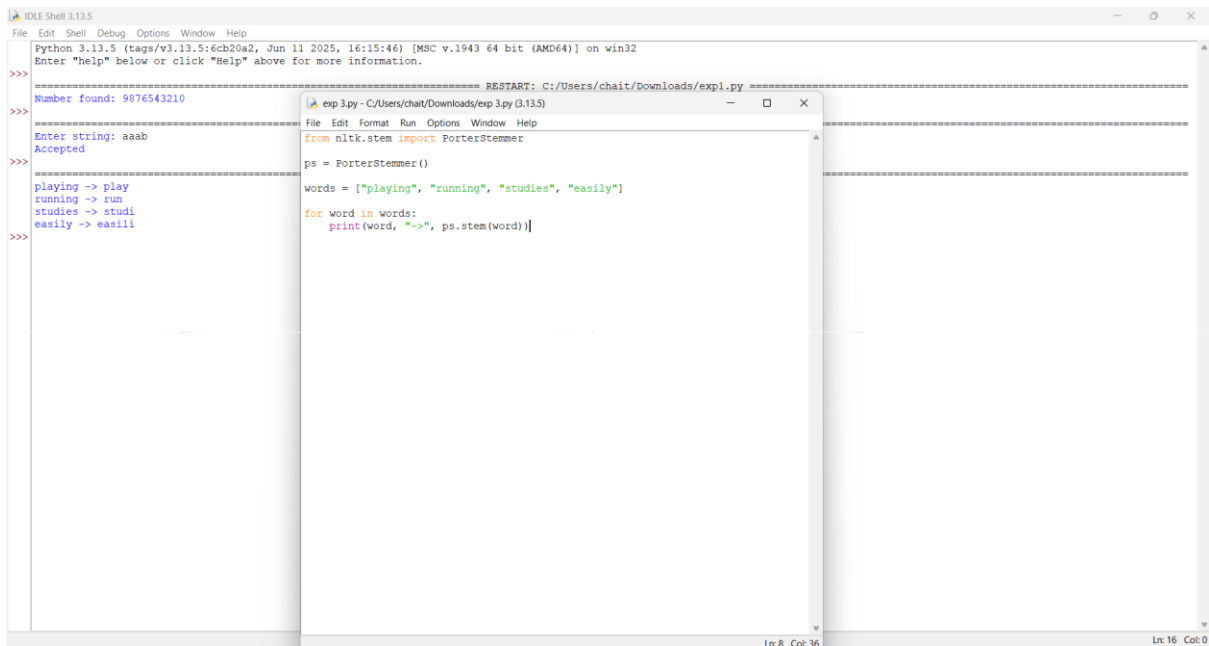
ps = PorterStemmer()

words = ["playing", "running", "studies", "easily"]

for word in words:

    print(word, "->", ps.stem(word))
```

OUTPUT:



The screenshot displays a Python IDE with two windows. The background window is a terminal shell showing the execution of a Python script. It starts with a prompt for a string, followed by the execution of the script which prints the stems of the words 'playing', 'running', 'studies', and 'easily'. The foreground window is a Python script editor showing the source code of the program. The code imports the PorterStemmer class from the nltk.stem module, creates a PorterStemmer object, and then iterates over a list of words, printing each word followed by its stem.

```
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
Number found: 9876543210
>>>
Enter string: aaab
Accepted
>>>
playing -> play
running -> run
studies -> studi
easily -> easili
>>>
```

```
exp3.py - C:/Users/chaith/Downloads/exp3.py (3.13.5)
File Edit Format Run Options Window Help
from nltk.stem import PorterStemmer

ps = PorterStemmer()

words = ["playing", "running", "studies", "easily"]

for word in words:
    print(word, "->", ps.stem(word))
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